



8. When a hash function is used to provide message authentication, the hash function value is referred to as
- A) Message Field B) Message Digest
C) Message Score D) Message Leap
9. Where is a Blockchain's central server?
- A) Where the blockchain is created B) Located with the owner of the server
C) There is no central server, it is distributed D) Ordered Node
10. If a Merkle tree has 8 transactions, how many hashes are required to compute the Merkle root?
- A) 8 B) 15
C) 16 D) 7
11. What is a nonce in the context of Bitcoin mining?
- A) The transaction ID number B) A miner's ASIC chip array
C) The generator point used in elliptic curve cryptography D) A value miners iterate through to generate a valid hash
12. Blockchain does not provide?
- A) Security B) Immutability
C) Fault tolerance D) Fast transaction time
13. Accounts in Ethereum are represented by
- A) hexadecimal address B) binary address
C) octal address D) network address
14. What is gas in the Ethereum context?
- A) The computational resources required to perform transactions. B) The security level of the Ethereum network.
C) The gas produced by Ethereum servers. D) The required cost to perform a transaction on the network.
15. The term used for a blockchain splits is ____.
- A) A merger B) A fork
C) A division D) None of the above
16. Which of the following statements is true about "inheritance" in Solidity?
- A) Inheritance allows a contract to borrow functionality and data from another contract. B) Inheritance is used to convert one data type to another.
C) Inheritance is a way to send Ether from one address to another. D) Inheritance is used to define the constructor of a smart contract.
17. What is the purpose of the fallback function in Solidity?
- A) To handle Ether transfers to the contract those do not match any function signature. B) To revert transactions if certain conditions are not met.
C) To deploy the contract on the Ethereum blockchain. D) To execute mathematical calculations within the contract.
18. What is a node?
- A) A type of cryptocurrency B) A Blockchain
C) A computer on a Blockchain network D) An exchange
19. What is Hyperledger?
- A) A cryptocurrency like Bitcoin B) A permissioned blockchain platform
C) A cloud computing service D) A programming language
20. In Hyperledger Fabric, Chaincode is equivalent to:
- A) A cryptocurrency mining algorithm B) A smart contract
C) A database system D) A security layer

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Describe the following terms and diagrammatically depict the relationship between them.				
	i) Public block chains				
	ii) Private block chains				
	iii) Hybrid blockchain				
	iv) Consortium blockchains	12	L*2	1	1
b)	List the benefits and limitations of block chain.	4	L2	1	1
2. a)	List and Explain the various features of blockchain.	6	L2	1	1
b)	Explain the five steps used to add a block to the blockchain.	5	L2	1	1
c)	Define Blockchain. With a suitable diagram discuss the generic structure of a block.	5	L2	1	1
3. a)	Compare Stream Ciphers and Block Ciphers.	6	L2	2	1
b)	Explain Merkle Tree concept used in Blockchain technology.	5	L2	2	1
c)	Write a note on digital signature.	5	L2	2	1
Unit – II					
4. a)	What is Recardian contract? List its properties.	6	L2	3	1
b)	Define a Smart contract and discuss the four important characteristics of smart contract.	6	L2	3	1
c)	Write short notes on:				
	i. Ethereum Stack				
	ii. Fork				
	iii. Gas	4	L2	3	1
5. a)	With a neat generic model explain the role and types of oracle in smart contract paradigm.	8	L2	3	1
b)	Explain the operation of Ethereum Virtual Machine (EVM) with a suitable diagram.	8	L2	3	1
6. a)	With suitable example, explain following in Solidity				
	i. Events				
	ii. Fall back functions				
	iii. Internal function call				
	iv. External function call				
	v. Modifier functions	10	L3	4	1
b)	With a simple example program, explain the layout of a solidity program.	6	L3	4	1
Unit – III					
7. a)	With a neat diagram, explain Hyperledger Fabric reference architecture.	8	L2	5	1
b)	Explain transaction processing flow in hyperledger fabric.	8	L2	5	1
8. a)	Explain advantages of Hyperledger and its family of projects to the industry.	6	L2	5	1
b)	Discuss various nodes involved in processing hyperledger transaction and various phases in carrying out a chaincode operation.	6	L2	5	1
c)	Write a note on four Hyperledger frameworks.	4	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
